



Sustainable Development of the Russian Border Regions: A Case Study of Key Goal Achievement in the Republic of Buryatia and Zabaikalsky Krai

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Abstract: The 2030 Agenda for Sustainable Development faces considerable implementation challenges in border regions, where natural and socio-economic processes transcend administrative boundaries. This study assesses progress towards selected Sustainable Development Goals (SDGs) in two Russian border regions: the Republic of Buryatia and Zabaikalsky Krai, which share borders with China and Mongolia, and identifies directions for bilateral cooperation based on the transboundary challenges. Based on official data from the Federal State Statistics Service of Russia for 2010–2024, the authors analysed 39 indicators covering SDGs 6, 8, 11, 13, 15 and 17. Growth rates were calculated and compared with national averages. The results reveal contrasting development models: Zabaikalsky Krai shows strong economic growth (gross regional product per capita index 114.3% in 2023) but critical deficits in access to quality drinking water (58.1% vs. national 89.2%), safe sanitation (49.2%), buses for persons with reduced mobility (3.2%), and protected forest areas (0.94%). The Republic of Buryatia performs better in social and environmental spheres (urban water access 86.0%, sanitation 80.5%, equipped buses 61.9%, reforestation ratio 247%) but lags economically—gross regional product (GRP) per capita index at 101.1%—and faces unique constraints from strict Baikal environmental norms (only 2.5% of wastewater meets those standards). Both regions suffer extreme wildfires and dispose of almost all municipal solid waste in landfills without recycling. The conclusions indicate that joint bilateral actions should include harmonised environmental standards and monitoring methodologies, creation of cross-border demonstration zones, and coordinated wildfire and water management. These findings can inform regional strategies, bilateral environmental projects, and statistical harmonisation within BRICS and the Shanghai Cooperation Organisation.

Keywords: Sustainable Development Goals; Border region; Russia; Republic of Buryatia; Zabaikalsky Krai; Environmental monitoring; Transboundary cooperation

1. Introduction

The adoption by the United Nations General Assembly in September 2015 of the resolution “Transforming our world: the 2030 Agenda for Sustainable Development” established a universal system of seventeen interlinked goals and 169 targets covering economic, social, and environmental dimensions of development (United Nations, 2015). This agenda resulted from a long evolution of the sustainable development concept, beginning with the report Our Common Future (World Commission on Environment & Development, 1987) and gaining momentum at the Rio de Janeiro Earth Summit (United Nations, 1992). However, practical implementation of the Sustainable Development Goals (SDGs) faces serious difficulties in border regions, where natural processes and economic activities are not confined by state borders.

A border region as a management object represents a territorial entity where three coordination systems intertwine: national (covering two or more countries), regional (represented by federal subjects and provinces), and local (including municipalities). Classical SDG monitoring approaches, focused primarily on the national level, prove insufficient for analysing and managing processes in such entities.

This issue is especially relevant in the Eurasian space, particularly at the interface of Russia with China and Mongolia. The length of the Russia-China state border is 4,209.3 km, and the Russia-Mongolia border is 3,485 km. More than 85 million people live in the border regions of Russia, China and Mongolia combined. These regions are characterised by high environmental vulnerability: here lie the basins of the transboundary rivers Amur, Argun and Selenga (the latter flowing through Mongolia and Russia into Lake Baikal), as well as the unique ecosystem of Lake Baikal, a United Nations Educational, Scientific and Cultural Organization (UNESCO) World Natural Heritage site. At the same time, there is significant asymmetry in socio-economic development: Chinese border provinces (Heilongjiang, Inner Mongolia) considerably surpass Russian regions in population and absolute gross regional product (GRP), but are inferior in GRP per capita. Mongolia, in turn, lags significantly behind both neighbours in economic terms, but its territory is an important part of the Selenga basin and an area of transboundary environmental risks.

In this study we analyse six SDGs that most depend on coordination of actions by neighbouring countries. The selection criteria were: physical connectivity of ecosystems (SDG 6—water resources, SDG 13—climate, SDG 15—biodiversity), economic interdependence (SDG 8—economic growth, SDG 11—urban development), and the possibility of institutional policy alignment (SDG 17—partnership). The objects of the study are the Republic of Buryatia and Zabaikalsky Krai—two border subjects of the Russian Federation directly bordering China and Mongolia and belonging to the Baikal region with a special environmental protection status.

The empirical analysis presented in this paper is limited to the two Russian regions described above. Mongolia and China are not included as empirical cases. Mongolia appears only as a geographical neighbour and as part of the transboundary context (for example, the Selenga basin and the Dauria reserve), but is excluded from the empirical analysis due to the lack of comparable statistical data for the 2010–2024 period. China, similarly, does not serve as a source of comparative statistical data; rather, it enters the discussion as an institutional reference point—drawing on its documented experience in SDG monitoring and demonstration zones—and as a partner for the bilateral actions proposed later in this paper. The authors acknowledge these limitations and intend to address them in future research.

The aim of this work is a comprehensive assessment of the level and dynamics of achieving these SDGs in the two regions based on official statistical data for 2010–2024, identification of systemic problems and disproportions, and justification of directions for joint actions, taking into account a comparative analysis of national monitoring systems and best practices of Russia and China. To achieve this goal, the following tasks were solved: analysis of time series of key SDG indicators for the border region; interpretation of the identified differences considering regional specifics (including the effect of special environmental norms on the Baikal Natural Territory); and development of practical recommendations for harmonising cross-border cooperation.

2. Literature Review

The literature on sustainable development in border and cross-border regions spans multiple disciplines, from environmental science and international relations to regional economics and statistics. To navigate this diverse body of knowledge, the present review is organised around three thematic inquiries. First, it examines what has been established regarding the challenges of SDG implementation and assessment in border and transboundary contexts globally. Second, it identifies the dominant methodological approaches employed in regional SDG assessment studies. Third, it delineates the specific gaps that remain unaddressed for the Russia-China-Mongolia borderland—in particular, the absence of harmonised indicator frameworks and empirical evidence on transboundary spillover effects. The following subsections address each of these questions in turn.

2.1 Sustainable Development Goal Challenges in Border and Cross-Border Regions

Border and cross-border regions face unique implementation challenges, including differences in national legislation, institutional barriers, cultural differences, and uneven economic development (Ferrer-Roca et al., 2020). These regions are often characterised by peripheral location, which complicates the achievement of sustainable development goals (Medeiros, 2020). At the same time, they share common natural systems, interdependent economies, and migration flows.

The physical connectivity of ecosystems determines the particular importance of SDGs 6, 13 and 15 for such regions. Water resources in transboundary river basins do not obey administrative boundaries; uncoordinated water use leads to conflicts, pollution and ecosystem degradation (Zeitoun & Warner, 2006). Climate risks (heat waves, floods, shifts in agro-climatic zones) are transboundary in nature; adaptation measures implemented in isolation can shift risks to adjacent areas (Intergovernmental Panel on Climate Change, 2023). Transboundary landscapes include migration corridors and forests critical for biodiversity; fragmentation of management disrupts ecological connections (Hilty et al., 2019). Without effective partnership (SDG 17), implementation of the other SDGs in a cross-border context is impossible, requiring multi-level governance from interstate agreements to local communities (Nilsson et al., 2016; Nilsson et al., 2018).

2.2 Methodological Approaches to Sustainable Development Goal Assessment at National and Regional Levels

National monitoring systems differ significantly. Russia's Federal State Statistics Service (Rosstat) approved a national list of 188 indicators, of which 124 are disaggregated to the level of federal subjects, creating an empirical basis for regional analysis (Tyrsin & Vasilyeva, 2020). However, the terms "sustainable development" and "green economy" were not officially fixed in Russian strategic planning documents for a long time, reducing implementation incentives (Bataeva & Kozhevina, 2020).

China's system is highly centralised and institutionalised. An Inter-agency Coordination Group for the Implementation of the 2030 Agenda was established in 2016, led by the Ministry of Foreign Affairs and the National Development and Reform Commission. China's National Bureau of Statistics developed more than 100 indicators adapted to the country's five-year plans (Zhu, 2017; Zhu et al., 2018). A key institutional feature is the linkage of SDG achievement to the performance evaluation system of local authorities (key performance indicators for officials), which creates powerful implementation incentives (Xie et al., 2021). China's experience in creating sustainable development demonstration zones—"Pilot Innovation Zones for the 2030 Agenda" in Guizhou, Guangdong, and Shaanxi provinces—follows a "piloting → verification → scaling" architecture that allows global goals to be adapted to local conditions (Wang et al., 2023; Wang et al., 2020).

Mongolia faces distinct challenges in SDG monitoring and implementation. The country's innovation development index (according to Cornell University and INSEAD methodology) shows that Mongolia is still in the process of forming a national innovation system (Filin et al., 2020). Quantitative analysis of the coupling coordination degree between urbanisation and the eco-environment in 21 Mongolian aimags revealed that most regions are at a stage of seriously unbalanced development; only the capital Ulaanbaatar achieved barely balanced development (index value 0.600 in 2016) (Dong et al., 2019). The methodology for disaggregated data collection and survey schemes that account for SDGs at sub-national and regional levels remains underdeveloped, particularly in mountain regions like the Russia-Mongolia borderland (Kulonen et al., 2019).

Methodological innovations include using the concept of "evenness" in addition to aggregate progress indicators and cloud model-based methods for monitoring urban land use under uncertainty (Wei et al., 2023). Recent bibliometric analysis confirms that research on SDG measurement has consolidated around three core clusters—the energy-emissions-growth nexus, renewable energy transitions, and econometric assessment methodologies—while persistent gaps include insufficient data disaggregation and a lack of integrated assessment models that capture interlinkages across SDGs (Siddiqui et al., 2023).

2.3 Gaps in the Russia-China-Mongolia Border Context

Despite significant progress, works devoted to the regional and provincial dimension of cross-border interactions, especially for the Russia-China-Mongolia triangle, are extremely rare. There are no agreed sets of indicators reflecting the specificities of this borderland, nor empirical studies quantitatively measuring the impact of changes in SDG indicators of one border territory on neighbouring ones.

Existing studies on the Russia-Mongolia border region reveal significant socio-economic asymmetries. The aggregate GRP of the two Russian border regions (the Republic of Buryatia and Zabaikalsky Krai) exceeds Mongolia's GDP by 36%, but Mongolia's average economic growth rate (8.4% in 2002–2012) significantly outpaced that of the Transbaikal regions (4%) (Atanov & Munkodugarova, 2016). About 90% of Mongolian exports consist of mineral resources, creating risks of "Dutch disease" and resource dependence—a problem also relevant to the Russian border regions (Atanov & Munkodugarova, 2016; Violin, 2018).

Environmental risks are substantial. In the Mongolia-China-Russia economic corridor, these include air pollution (particularly from electricity production and distribution), groundwater pollution, flooding, waste accumulation, and land degradation (Dong et al., 2018). In Mongolia specifically, desertification, soil erosion, and pastureland degradation have been identified as critical issues (Filin et al., 2020). The protection of Lake Baikal—a UNESCO World Heritage site—imposes obligations on both countries, including refraining from any deliberate actions that could directly or indirectly damage the heritage (Bychkov & Orlova, 2018). Mongolia's plans to construct hydroelectric facilities in the Selenga basin (the main tributary of Baikal) have been subject to active international consultations, with the World Heritage Committee since 2012 requiring that no projects be approved until environmental impact assessments are completed (Bychkov & Orlova, 2018).

A Delphi-based study in the Cambodia-Laos-Vietnam (CLV) transboundary area demonstrates the feasibility of harmonising indicators across three countries. Starting from 197 potential indicators, the authors reduced these to 70 key indicators (24 for land, 19 for water, 27 for forest resources) using a combination of focus group discussions and a two-round Expert Delphi process, achieving strong expert consensus (Kendall's $W = 0.50$) (Alsaif et al., 2025). This experience highlights the value of structured, participatory methodologies for harmonising indicators across national borders—a lesson directly applicable to the Russia-China-Mongolia

context, where Rosstat, China's National Bureau of Statistics and Mongolia's National Statistics Office operate under different frameworks.

Thus, while Russia, China and Mongolia possess relevant institutional and methodological experience—regional indicator disaggregation in Russia, demonstration zones and performance-linked monitoring in China, and an emerging innovation system in Mongolia—no agreed framework for tripartite SDG monitoring exists. The practical mechanisms for harmonising indicators, data standards and governance practices across this borderland have yet to be developed. This study contributes to filling this gap by providing an empirical assessment of two Russian border regions—the Republic of Buryatia and Zabaikalsky Krai—and outlining directions for trilateral cooperation between Russia, China and Mongolia in the field of environmental coordination and SDG monitoring.

3. Methodology

3.1 Study Area

The study covers two subjects of the Russian Federation located in the southern part of Eastern Siberia and sharing a common border with Mongolia and China (Figure 1). The Republic of Buryatia lies between 50–57° N and 98–110° E. Its area is 351,300 square kilometres, most of which lies within the Baikal Natural Territory with a special natural resource use regime. More than 80% of the republic's area is occupied by mountain ranges and highlands (Khamar Daban, Barguzinsky, Eastern Sayan). The Republic of Buryatia possesses a unique hydrological feature—Lake Baikal, which is the largest freshwater reservoir on the planet and is included in the UNESCO World Natural Heritage list.

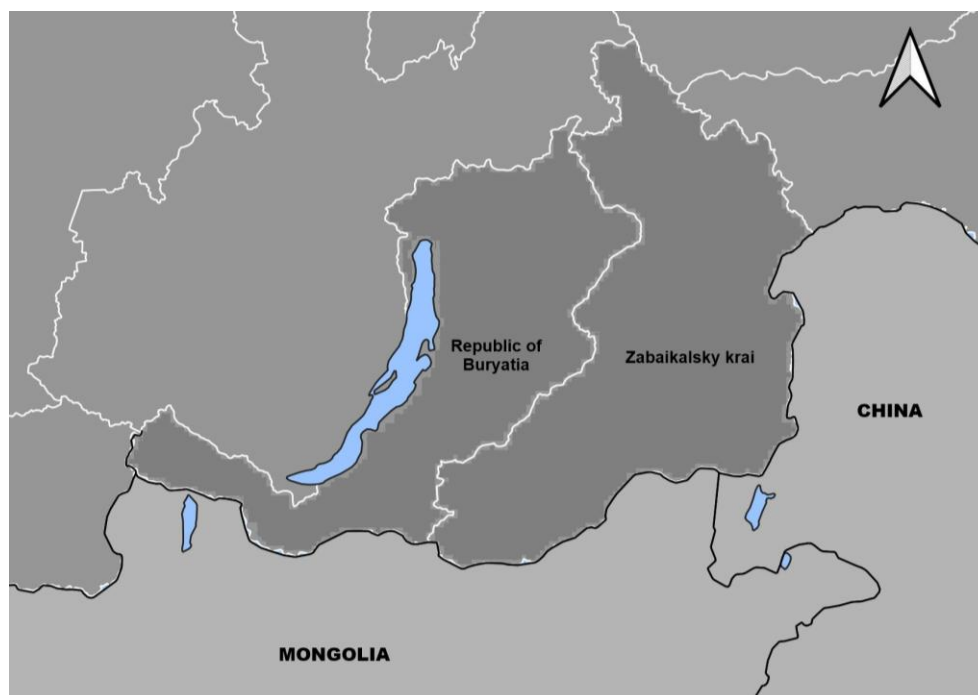


Figure 1. The study area

Zabaikalsky Krai, formed in 2008 through the merger of Chita Oblast and Agin Buryat Autonomous Okrug, is located between 49–58° N and 108–123° E. Its area is 431,900 square kilometres. The region is characterised by a complex relief with a predominance of mid mountains and intermontane basins. The climate of both regions is sharply continental, with long cold winters, short warm summers, and extremely uneven precipitation distribution, which creates elevated fire danger.

The choice of these regions for analysis is based on several factors. First, their border location makes them most vulnerable to transboundary environmental risks and simultaneously most promising for the development of bilateral economic cooperation. Second, the presence of the unique Baikal ecosystem and the Amur basin requires special attention to water and forest resources. Third, both regions exhibit pronounced socio-economic disproportions typical of many Russian border areas. Fourth, joint projects already exist here—for example, the transboundary reserve “Dauria” (Russia, China, Mongolia)—which creates an institutional basis for broader cooperation in the field of sustainable development.

3.2 Materials and Methods

The study uses official data from the Federal State Statistics Service of the Russian Federation (Rosstat) for the period from 2010 to 2024. In total, 39 indicators covering a time interval sufficient to identify stable trends were analysed. The selection of six SDGs and 39 indicators followed a three-step screening process: (a) theoretical relevance to transboundary interactions (connectivity of ecosystems for SDGs 6, 13 and 15; economic interdependence for SDG 8; spatial planning for SDG 11; institutional alignment for SDG 17); (b) data availability for both regions; and (c) alignment with Rosstat's official regional SDG monitoring list. The actual coverage period varies by indicator; Table 1 specifies the exact time range for each indicator.

Table 1. Indicators for assessing the achievement of key Sustainable Development Goals (SDGs) in the border region

No.	Indicator Name	Unit of Measurement	Period/Coverage
<i>SDG 6—Clean Water and Sanitation</i>			
6.1	Share of the population with access to quality drinking water from centralized water supply systems—total population	%	2018–2024, annual
6.2	Share of urban population with access to quality drinking water from centralized water supply systems	%	2018–2024, annual
6.3	Share of households with access to centralized water supply	%	2014, 2016, 2018, 2020, 2022, 2024
6.4	Share of the population using safely managed sanitation services (including handwashing facilities with soap and water)	%	2018, 2020, 2022, 2024
6.5	Share of wastewater treated to national standards in total wastewater requiring treatment	%	2010–2024, annual
<i>SDG 8—Decent Work and Economic Growth</i>			
8.1	Index of physical volume of gross regional product per capita	% of previous year	2010–2023, annual
8.2	Labour productivity index	% of previous year	2012–2023, annual
8.3	Employment rate (population aged 15 years and over)	%	2017–2024, annual
8.4	Composite indicator of unemployment and potential labour force	%	2017–2024, annual
8.5	Share of youth aged 15–24 not in education, employment or training (NEET)	%	2017–2024, annual
8.6	Share of employees in organisations (excluding small enterprises) with wages below the subsistence minimum	%	2011, 2013, 2015, 2017, 2019, 2021, 2023, 2025 (forecast)
8.7	Number of injured with work disability (≥ 1 day) and fatal injuries per 1,000 employed persons	persons per 1,000 employed	2010–2024, annual
8.8	Share of gross value added of the tourism industry in gross regional product	%	2019–2023, annual
<i>SDG 11—Sustainable Cities and Human Settlements</i>			
11.1	Share of urban population living in dilapidated housing stock	%	2019–2024, annual
11.2	Share of households experiencing overcrowding	%	2014, 2016, 2018, 2020, 2022, 2024
11.3	Share of cities with a favourable urban environment (urban quality index $>50\%$)	%	2018–2023, annual
11.4	Number of citizens relocated from unfit housing stock	thousand persons	2018–2024, annual
11.5	Share of operational buses equipped for transporting persons with reduced mobility	%	2011–2024, annual
11.6	Share of urban agglomeration road network in good condition	%	2021–2024, annual
11.7	Volume of pollutant emissions from road transport	thousand tonnes	2019–2024, annual
11.8	Budget allocations for conservation of cultural heritage sites	million rubles	2010–2024, annual
11.9	Share of captured and neutralized air pollutants from stationary sources in total pollutants generated	%	2018–2024, annual
11.10	Share of length of lit urban streets, driveways, embankments in total length of urban streets	%	2010–2024, annual
11.11	Share of green areas within city boundaries in total urban land area	%	2010–2024, annual
11.12	Share of municipal solid waste sent to landfill (including waste that has been sorted) in total MSW generated	%	2020–2024, annual
11.13	Ratio of the rate of housing construction to the rate of population growth	coefficient	2024 (calculated year)
<i>SDG 13—Climate Action</i>			
13.1	Area of forest land affected by fires	hectares	2010–2024, annual
13.2	Area of non-forest land affected by fires	hectares	2010–2024, annual
13.3	Number of sectoral and regional climate change adaptation plans	units	2021–2024, annual

<i>SDG 15—Life on Land</i>			
15.1	Forest area as percentage of total land area (forest cover)	%	2010–2024, annual
15.2	Share of protected areas (federal, regional, local) in total area	%	2014–2024, annual
15.3	Share of forest area within legally established protected areas	%	2024 (single year)
15.4	Ratio of reforestation and afforestation area to area of logged and dead forest stands	%	2018–2024, annual
15.5	Index of physical volume of environmental expenditure on biodiversity conservation and protection of natural areas	% of previous year	2019–2024, annual
<i>SDG 17—Partnerships for the Goals</i>			
17.1	Total government revenue as percentage of gross domestic product	%	2010–2024, annual
17.2	Gross regional product per capita (current prices)	rubles	2010–2023, annual
17.3	Index of physical volume of GRP per capita (constant prices)	% of previous year	2010–2023, annual
17.4	Share of population using the Internet (aged 15–74 years)	%	2014–2024, annual
17.5	Dollar value of all resources allocated to building statistical capacity of developing countries	million US dollars	2017–2024, annual

Note: GRP = gross regional product.

For each indicator, absolute values for the last available year (mostly 2023 or 2024) and base growth rates for the period 2010–2024 (where data permitted) were calculated. Comparison with all Russian indicators allowed us to assess the degree of deviation of the regions from the national average. For dynamics analysis, chain indices (percentage of the previous year) were used, allowing the identification of periods of acceleration or deceleration of progress.

Any potential discontinuities in regional statistical time series (e.g., due to changes in reporting standards or external shocks such as the COVID-19 pandemic) are implicitly accounted for by comparing regional dynamics with the all-Russian average, as methodological changes affect regional and national data in a consistent manner.

To provide a consolidated synthesis of regional performance across the six SDGs, the authors developed a transparent two-level aggregation procedure. At the first level, each of the 39 indicators was classified into one of four categories based on its deviation from the Russian average. An indicator was rated as *Above* if its regional value exceeded the national average by more than 5%, and as *Below* if it fell below the national average by more than 5%. Values falling within $\pm 5\%$ of the national average were considered *Near average*. The *Critically below* category was reserved for indicators where the regional value fell below the national average by more than 30 percentage points, or for extreme anomalies that fundamentally distort the assessment regardless of the percentage gap. At the second level, these indicator-level classifications were aggregated to the SDG level. A rating of *Above* was assigned when the majority of indicators for a given SDG fell into the *Above* category with no indicators rated as *Critically below*. A rating of *Below* was given when the majority of indicators were classified as *Below*, with no more than one indicator in the *Critically below* category. The *Critically below* rating was used when two or more indicators fell into that category, or when at least one indicator represented an extreme anomaly. A *Mixed* rating was applied when performance was divergent, with no clear majority in any single category—for example, when some indicators were rated *Above* while others fell into *Below* or *Critically below*. The *Near average* rating was used when the majority of indicators fell within the *Near average* category with no indicators rated as *Critically below*. An additional category, *Extreme vulnerability*, was introduced to capture situations where a region experiences an event or trend of exceptional severity that significantly exceeds national averages in absolute terms, shows anomalous dynamics, poses risks of irreversible environmental or socio-economic damage, and may have transboundary implications. This category is applied irrespective of comparison with the Russian average when the scale and dynamics of the phenomenon are objectively extreme, regardless of which SDG is affected.

In addition to quantitative analysis of statistical data, the study includes a comparative analysis of the national SDG monitoring systems of Russia and China based on regulatory and literature sources. China and Mongolia are not included as empirical cases in the quantitative part of this study due to data comparability constraints; they serve as institutional reference points for the discussion of bilateral and trilateral cooperation.

4. Results

The analysis of statistical data revealed significant differences between the Republic of Buryatia and Zabaikalsky Krai for most of the analysed indicators, as well as a considerable lag of both regions behind all-Russian averages for many socio-environmental parameters, while simultaneously showing an advance for certain economic indicators.

In the area of access to clean water and sanitation (SDG 6), the following results were obtained (Figure 2). The share of the population of the Russian Federation with access to quality drinking water from centralized water supply systems in 2024 was 89.2%. In the Republic of Buryatia this indicator reached only 56.3%, and in Zabaikalsky Krai—58.1%. Thus, the gap from the national level exceeds 30 percentage points, placing these regions among the most problematic for this indicator. However, when considering the urban population, the

situation is substantially better: 86.0% of urban residents in Buryatia and 69.2% of urban residents in Zabaikalsky Krai have access to quality water (compared to 95.4% in Russia). This indicates that the main water supply problems are concentrated in rural areas and small towns.

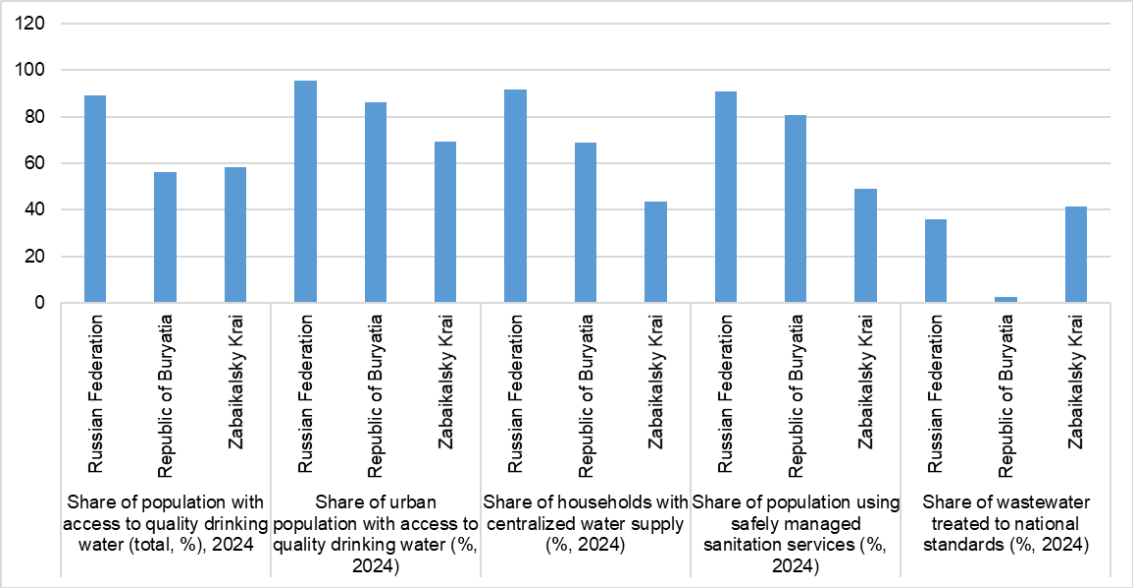


Figure 2. Assessment of Sustainable Development Goal (SDG) 6 achievement indicators

The share of households with access to centralized water supply in the Russian Federation is 91.7%. In Buryatia this indicator is 68.9% (which, nevertheless, demonstrates significant progress: growth from 50.7% in 2014 to 68.9% in 2024). In Zabaikalsky Krai the situation is considerably worse—only 43.7% of households have centralized water supply, with an increase of only 11 percentage points over the decade. An even more acute situation concerns the share of the population using safely managed sanitation services (including handwashing with soap and water). In Zabaikalsky Krai this indicator in 2024 was 49.2%—one of the lowest in the country, comparable only to the Republic of Tyva (9%) and the Chechen Republic (29%). In Buryatia the situation is better—80.5%—but still lags behind the national average (90.9%).

The indicator “Share of wastewater treated to national standards” presents a paradoxical picture. On average in the Russian Federation the share increased from 10.2% in 2010 to 35.8% in 2024. In Zabaikalsky Krai this indicator stands at 41.6%, which is even somewhat above the national average. In Buryatia, however, the share of treated wastewater is only 2.5%—14 times lower than the Russian average. As has been established, this anomalously low value is not a consequence of the absence of treatment facilities or their extreme depreciation. The Republic of Buryatia is located within the Baikal Natural Territory, which is subject to Order No. 63 of the Ministry of Natural Resources and Ecology of the Russian Federation dated 5 March 2010 “On approval of the standards for maximum permissible impacts on the unique ecological system of Lake Baikal and the list of hazardous substances, including substances classified as especially hazardous, highly hazardous, hazardous and moderately hazardous for the unique ecological system of Lake Baikal”, establishing maximum permissible concentrations of pollutants that are tens of times stricter than all-Russian standards. The existing treatment facilities, built in the 1970s–1980s, are physically incapable of achieving such a degree of purification. New water treatment plants capable of operating under the Baikal standards require the introduction of ozonation and membrane filtration technologies, which multiplies the cost of projects. Therefore, the entire volume of wastewater passed through outdated treatment facilities formally does not meet the Baikal requirements, which is reflected in the statistics. In Zabaikalsky Krai, where the main waterways belong to the Amur basin, such ultra-strict requirements are not applied.

In the area of economic growth and decent work (SDG 8), the picture is also heterogeneous (Figure 3). The index of physical volume of gross regional product per capita in Zabaikalsky Krai in 2023 was 114.3% of the previous year’s level, whereas the Russian average was 104.4%. In Buryatia growth was more modest—101.1%. An even more striking gap is observed in labour productivity: the labour productivity index in Zabaikalsky Krai in 2023 reached 116.0% (compared to 102.3% in Russia), while in Buryatia a decline was recorded—97.5%. Absolute GRP per capita at current prices in 2023 was 726.8 thousand rubles in Zabaikalsky Krai, 517.8 thousand rubles in Buryatia, and the Russian average—1,073.7 thousand rubles. Thus, Zabaikalsky Krai attains 68% of the national average for this indicator, while Buryatia reaches only 48%.

Employment and unemployment indicators reveal a more complex picture. The employment rate of the

population aged 15 years and over in 2024 was 61.4% in the Russian Federation. In Buryatia this indicator was 54.1%, and in Zabaikalsky Krai—57.3%. The composite unemployment indicator, which takes into account not only the officially registered unemployed but also persons in the potential labour force (those wishing to work but not seeking work or not ready to start), stood at 3.5% in Russia, 9.9% in Buryatia, and 7.0% in Zabaikalsky Krai in 2024. Of particular concern is the share of youth aged 15–24 not in education, employment, or training (NEET). In 2024 this indicator was 8.4% in Russia, 14.8% in Buryatia, and 9.9% in Zabaikalsky Krai. Positively, Zabaikalsky Krai experienced a sharp decline from 16.5% in 2017, whereas in Buryatia the dynamics have been negligible.

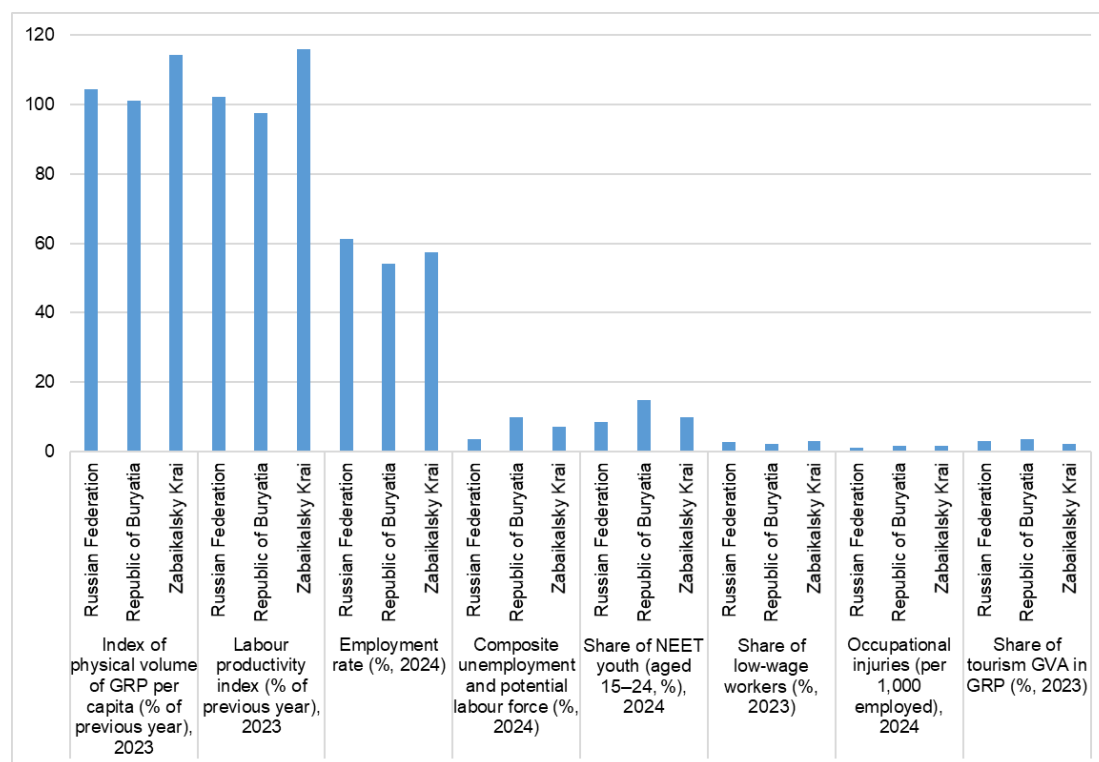


Figure 3. Assessment of Sustainable Development Goal (SDG) 8 achievement indicators

Occupational injuries in both regions remain above the national average. The number of injured with work disability per 1,000 employed persons in 2024 was 1.0 in Russia, 1.7 in Buryatia, and 1.6 in Zabaikalsky Krai. Moreover, in Zabaikalsky Krai an increase in injuries was recorded compared to 2023 (from 1.3 to 1.6), which may be associated with the intensification of industrial production. The share of gross value added of the tourism industry in GRP in 2023 was 2.9% in Russia, 3.54% in Buryatia, and only 2.16% in Zabaikalsky Krai. The Baikal tourism potential of Buryatia has not yet been fully realised—after the pandemic-induced decline in 2020 (3.4%), the indicator has not recovered to its pre-crisis level of 4.56% (2019).

In the sphere of sustainable cities and human settlements (SDG 11), both positive and negative trends were identified (Table 2). The share of the urban population living in dilapidated housing stock in 2024 was 0.51% in Russia, 1.17% in Buryatia, and 1.67% in Zabaikalsky Krai. Thus, in Zabaikalsky Krai this indicator is more than three times higher than the national average. The relocation of citizens from dilapidated housing is proceeding slowly: in 2024, 2.91 thousand persons were relocated in Zabaikalsky Krai, and 7.10 thousand in Buryatia (for comparison: in Russia—810.6 thousand persons). At the same time, the scale of dilapidated housing in both regions is growing or stabilising at a high level.

The quality of the urban environment, measured by the share of cities whose index exceeds 50%, reached 68% in the Russian Federation, 33% in Buryatia, and 20% in Zabaikalsky Krai in 2023. The dynamics are positive (in Buryatia from 0% in 2018), but the initial level was catastrophically low. A particularly striking contrast is observed in the accessibility of public transport for persons with reduced mobility. The share of operational buses equipped for transporting disabled persons in 2024 was 44.3% in the Russian Federation. In Buryatia this indicator reached 61.9% (above the national average, with growth from 5% in 2020). In Zabaikalsky Krai, however, the share of equipped buses is only 3.2% and even decreased compared to 2022 (7.6%). This is one of the lowest rates in the country, making the transport system of Zabaikalsky Krai practically inaccessible for people with limited mobility.

Table 2. Assessment of Sustainable Development Goal (SDG) 11 achievement indicators

No.	Indicator Name	Territory	Value (Latest Year)	Dynamics
11.1	Share of urban population living in dilapidated housing (% , 2024)	Russian Federation	0.51	↓
		Republic of Buryatia	1.17	↑
		Zabaikalsky Krai	1.67	↑
11.2	Share of households experiencing overcrowding (% , 2024)	Russian Federation	15.0	↓
		Republic of Buryatia	18.1	↓
		Zabaikalsky Krai	11.1	↓
11.3	Share of cities with favourable urban environment (index >50%, %), 2023	Russian Federation	68	↑
		Republic of Buryatia	33	↑
		Zabaikalsky Krai	20	↑
11.4	Citizens relocated from dilapidated housing (cumulative since 2018, thousand persons), 2024	Russian Federation	810.6	↑
		Republic of Buryatia	7.10	↑
		Zabaikalsky Krai	2.91	↑
11.5	Share of buses equipped for persons with reduced mobility (% , 2024)	Russian Federation	44.3	↑
		Republic of Buryatia	61.9	↑
		Zabaikalsky Krai	3.2	↓
11.6	Share of urban agglomeration road network in good condition (% , 2024)	Russian Federation	85.7	↑
		Republic of Buryatia	83.5	↑
		Zabaikalsky Krai	87.3	↑
11.7	Pollutant emissions from road transport (thousand tonnes, 2024)	Russian Federation	4,839.1	↓
		Republic of Buryatia	38.1	↓
		Zabaikalsky Krai	27.0	↑
11.8	Budget allocations for cultural heritage conservation (million rubles, 2024)	Russian Federation	416,320	↑
		Republic of Buryatia	315	→
		Zabaikalsky Krai	71	→
11.9	Share of captured and neutralized air pollutants from stationary sources (% , 2024)	Russian Federation	73.4	→
		Republic of Buryatia	86.5	→
		Zabaikalsky Krai	81.9	↑
11.10	Share of lit urban streets (% , 2024)	Russian Federation	73.4	↑
		Republic of Buryatia	64.5	↑
		Zabaikalsky Krai	31.9	↑
11.11	Share of green areas within city boundaries (% , 2024)	Russian Federation	26.0	→
		Republic of Buryatia	22.2	→
		Zabaikalsky Krai	15.6	→
11.12	Share of municipal solid waste sent to landfill (% , 2024)	Russian Federation	82.5	↑
		Republic of Buryatia	100.0	↑
		Zabaikalsky Krai	99.7	→
11.13	Ratio of housing construction rate to population growth rate (coefficient, 2024)	Russian Federation	0.98	→
		Republic of Buryatia	1.14	↑
		Zabaikalsky Krai	1.20	↑

Note: ↑ = improvement (increase of a desirable indicator or decrease of an undesirable one); ↓ = deterioration (decrease of a desirable indicator or increase of an undesirable one); → = stability/negligible change (less than 2% relative change or 1 percentage point absolute change).

The condition of the road network in urban agglomerations has noticeably improved in Zabaikalsky Krai: the share of roads in good condition increased from 64.8% in 2021 to 87.3% in 2024, exceeding the national average (85.7%). In Buryatia this indicator also increased (from 79.0% to 83.5%), remaining close to the average. Pollutant emissions from road transport in Buryatia decreased from 2019 to 2023 (from 39.9 to 37.2 thousand tonnes), but in 2024 rose slightly to 38.1 thousand tonnes. In Zabaikalsky Krai, by contrast, emissions increased from 24.6 thousand tonnes in 2020 to 27.0 thousand tonnes in 2024, indicating a negative trend.

The share of captured and neutralised air pollutants from stationary sources in 2024 was 73.4% in the Russian Federation, 86.5% in Buryatia, and 81.9% in Zabaikalsky Krai. This indicates good equipment of industrial enterprises with treatment facilities, which is especially important given the high environmental vulnerability of the Baikal region. However, street lighting remains a problem: in the Russian Federation—73.4%, in Buryatia—64.5% (with steady growth from 42.8% in 2010), in Zabaikalsky Krai—only 31.9% (with slight growth from 24.1%). Low lighting creates road safety risks and reduces the comfort of the urban environment.

Green spaces within city boundaries occupy on average 26.0% of the area in the Russian Federation. In Buryatia this indicator varies around 22% (22.2% in 2024), in Zabaikalsky Krai—a consistently low 15.6% in recent years, significantly below the national average. The most alarming situation concerns municipal solid waste management. The share of municipal solid waste sent to landfill in total waste generated in the Russian Federation is 82.5% (in 2024). In Buryatia this indicator reached 100%, in Zabaikalsky Krai—99.7%. Effectively, all waste in these regions is taken to landfills and disposed of without any sorting or recycling, contradicting the principles

of a circular economy and creating a long-term environmental burden.

In the area of climate action (SDG 13), the most significant data concern forest fires. The total area of forest land affected by fires in the Russian Federation in 2024 amounted to 6.976 million hectares. Of this, the Republic of Buryatia accounted for 397 thousand hectares, and Zabaikalsky Krai—1.949 million hectares. Thus, in Zabaikalsky Krai about 28% of all forests destroyed by fires in Russia were burned. Particularly alarming is the dynamics: compared to 2023, the fire area in Buryatia increased 21-fold (from 18.5 thousand ha), and in Zabaikalsky Krai—33-fold (from 59.4 thousand ha). Non-forest lands also suffered from fire: in the Russian Federation—1.279 million hectares, in Buryatia—7.29 thousand ha, in Zabaikalsky Krai—37.87 thousand ha. The main causes of such large-scale fires are climatic anomalies—abnormal heat, drought, and strong winds—which confirms the high vulnerability of the region to climate change. Both regions have approved climate change adaptation plans; however, their effectiveness under the extreme fire conditions of 2024 proved insufficient.

In the sphere of terrestrial ecosystem conservation (SDG 15), the following data were obtained (Figure 4). Forest cover (the share of forest land in total land area) averages 52.8% in the Russian Federation, 72.3% in Buryatia, and 72.4% in Zabaikalsky Krai. Both regions are among the most forested in the country. The share of specially protected natural areas of federal, regional, and local significance in total territory in 2024 reached 14.3% in the Russian Federation. In Buryatia this indicator is stable at about 9.0% (with minor fluctuations); in Zabaikalsky Krai it increased from 5.4% in 2014 to 8.6% in 2024. Despite this increase, both regions lag behind the national average. The situation is particularly critical regarding the share of forest area within protected areas. For the Russian Federation this indicator is 2.22%. In Buryatia it is 6.31% (due to the presence of large federal protected areas—the Baikal Nature Reserve, the Barguzinsky Nature Reserve, and the Zabaikalsky National Park). In Zabaikalsky Krai, however, the share of forests under protection is only 0.94%—one of the lowest rates in the country. This means that large forest tracts susceptible to fires and logging remain outside the special protection system.

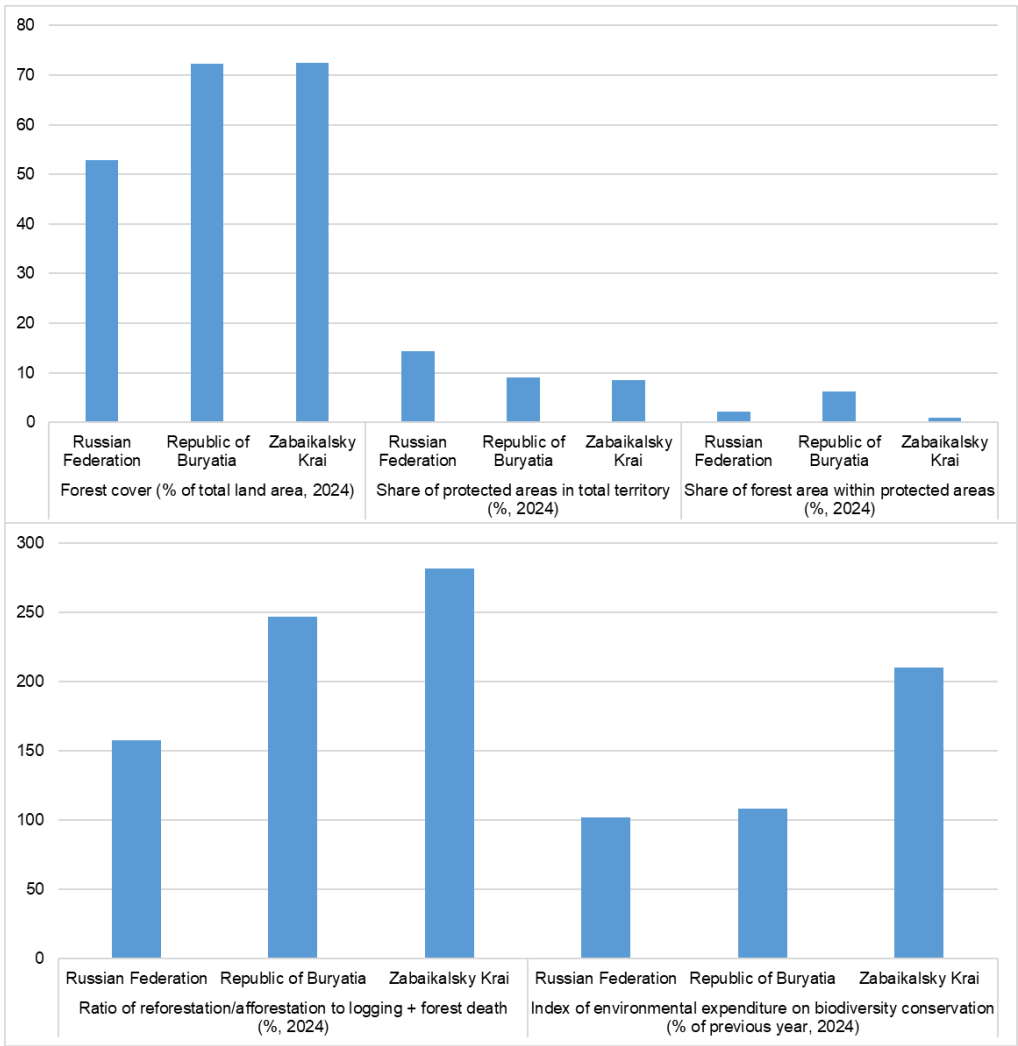


Figure 4. Assessment of Sustainable Development Goal (SDG) 15 achievement indicators

Reforestation is carried out very actively in both regions. The ratio of reforestation and afforestation area to the area of logged and dead forest stands in 2024 was 158% for the Russian Federation, 247% for Buryatia, and 282% for Zabaikalsky Krai. The dynamics are positive: in Zabaikalsky Krai the indicator increased from 27.9% in 2018 to 282% in 2024; in Buryatia—from 150.9% to 247%. Part of these high values is explained by natural reforestation (natural regrowth) on burnt areas and a time lag between forest mortality and its accounting; nevertheless, overall, they indicate considerable efforts. The index of physical volume of environmental expenditure on biodiversity conservation in 2024 was 101.8% for the Russian Federation, 108.4% for Buryatia, and 210% for Zabaikalsky Krai (i.e., a doubling compared to 2023). This indicates a growing priority of environmental protection activities.

In the sphere of partnerships and means of implementation (SDG 17), key indicators have already been partially presented in the economic growth section. Additionally, it should be noted that Internet access for the population aged 15–74 years in 2024 reached 94.4% in the Russian Federation, 97.3% in Buryatia, and 92.0% in Zabaikalsky Krai. High digital accessibility creates a foundation for the development of e-government, digital partnerships, and joint information monitoring systems. In 2024, the Russian Federation allocated USD 2.86 million to building statistical capacity in developing countries. Although these funds are not directly directed towards Russia–China cooperation, they indicate the existence of mechanisms for international support of statistical systems.

The summary assessment presented in Table 3 synthesises the performance of the two regions across the six SDGs analysed.

Table 3. Summary assessment of key Sustainable Development Goal (SDG) achievement in the Republic of Buryatia and Zabaikalsky Krai

SDG		Republic of Buryatia	Zabaikalsky Krai
6	Clean water and sanitation	Mixed	Critically below
8	Decent work and economic growth	Below	Above
11	Sustainable cities and settlements	Mixed	Critically below
13	Climate action	Extreme vulnerability	Extreme vulnerability
15	Life on land	Above	Below
17	Partnerships for the goals	Above	Near average

The assessment reveals contrasting sustainability profiles of the two regions. Zabaikalsky Krai demonstrates above-average economic performance (SDG 8) but is rated as critically below in water and sanitation (SDG 6) and sustainable cities (SDG 11), as well as below average in life on land (SDG 15). The Republic of Buryatia, by contrast, performs above the national average in life on land (SDG 15) and partnerships (SDG 17), shows mixed results in water and sanitation (SDG 6) and sustainable cities (SDG 11), and lags in economic growth (SDG 8). Both regions share extreme vulnerability under climate action (SDG 13) due to unprecedented forest fire areas in 2024. These findings indicate that each region faces a distinct combination of sustainability challenges, while transboundary environmental risks—particularly wildfires and water management in the Selenga basin—affect both regions and therefore require coordinated action.

5. Discussion

The results obtained allow several conclusions about the nature of sustainable development in the two Russian border regions and help identify factors explaining the observed disproportions.

The contrast between high economic growth rates in Zabaikalsky Krai and its lag on social and environmental indicators can be explained by the structure of the region’s economy. Zabaikalsky Krai is dominated by extractive industries (gold, uranium, molybdenum, coal) and transport infrastructure (the Baikal-Amur Mainline and Trans-Siberian Railway). These sectors provide rapid value-added growth but create a limited number of high-skill jobs while generating significant environmental loads. Labour productivity growth, combined with stagnant employment and a steadily declining population, leads to a situation where formal economic indicators improve, but the real well-being of broad sections of the population does not necessarily increase. Social indicators—unemployment, not in education, employment, or training (NEET) youth, dilapidated housing—reflect this situation. This pattern is consistent with the resource-dependent development model documented for the Transbaikal regions, where extractive industries drive economic output while failing to diversify employment opportunities (Atanov & Munkodugarova, 2016; Violin, 2018). In the Republic of Buryatia, by contrast, social infrastructure is better developed, thanks to federal programmes and a relatively more diversified economy with a noticeable share of tourism and small business. However, economic growth here is slower, possibly due to stricter environmental restrictions (the Baikal factor) and a smaller volume of investment in the extractive sector. The wastewater treatment problem in Buryatia has no simple technical solution under current regulations; either revision of the norms or the construction of treatment facilities using best available technologies is required. This regulatory constraint reflects the broader challenge of balancing economic development with the preservation of

the Baikal ecosystem, as previously highlighted in studies on transboundary water governance in the Selenga basin (Bychkov & Orlova, 2018).

Both regions face a set of shared problems with pronounced transboundary dimensions. The most acute is extreme forest fire susceptibility. In 2024, Zabaikalsky Krai lost nearly 2 million hectares of forest, accounting for about 28% of all forest fires in the Russian Federation. The reasons include not only anomalous weather conditions but also insufficient funding for firefighting measures, weak aerial forest protection, and potential fire transfer from neighbouring territories (China and Mongolia), although the current dataset does not allow quantification of this transboundary share. This directly suggests that a joint monitoring and early warning system could be a valuable tool. Environmental risks of this nature—including air pollution, groundwater contamination, flooding, waste accumulation, and land degradation—have been identified as critical challenges in the Mongolia-China-Russia economic corridor (Dong et al., 2018). In Mongolia specifically, desertification, soil erosion, and pastureland degradation compound these risks, underscoring the need for coordinated transboundary environmental management (Filin et al., 2020). A common problem for both regions is the near-total landfilling of municipal solid waste without recycling, which contradicts circular economy principles and creates a long-term environmental burden. Other shared issues include inadequate street lighting (particularly acute in Zabaikalsky Krai), a deficit of green spaces within urban boundaries, and persistent gaps in providing quality drinking water to rural populations. Such problems—fire propagation, water pollution, and contaminant migration—cannot be resolved unilaterally and necessitate coordinated international action.

A comparative analysis of the national SDG monitoring systems in Russia and China reveals that China's experience offers valuable lessons for Russian border regions. China's system is more detailed, centralised, and rigidly linked to official performance evaluation. The creation of demonstration zones—"Pilot Innovation Zones for the 2030 Agenda"—allows new approaches to be tested at the local level and then scaled up. Transferring this experience to the Russia-China border area could be achieved through the creation of joint "pilot sustainable development zones" where agreed indicators, monitoring methods, and coordination mechanisms would be tested. Such zones could be established in transboundary reserves (e.g., the Dauria reserve, which already involves Russia, China and Mongolia), transboundary river basins (the Selenga or Amur), or corridors of international transport routes. The experience of the CLV Development Triangle, where a Delphi-based approach successfully harmonised 70 indicators across three countries (Alsaif et al., 2025), provides a replicable template.

The findings of this study support the need for coordinated actions at different time horizons. In the short term, practical steps should include the harmonisation of environmental standards for key transboundary indicators (water quality in the Selenga basin, air quality in border cities) and the creation of a joint early warning system for forest fires, building on existing satellite monitoring capabilities. In the medium term, institutional arrangements should focus on the development of agreed monitoring methodologies, data sharing protocols, and joint environmental impact assessments for major infrastructure projects. In the long term, the establishment of trilateral (Russia-China-Mongolia) sustainable development demonstration zones could serve as platforms for testing integrated approaches to green growth, circular economy, and low-carbon development. These zones could also function as living laboratories for achieving SDG 17 (partnerships) while addressing the specific transboundary challenges identified in this study. The spatial dimension of these challenges is particularly relevant, as mountain regions like the Russia-Mongolia borderland face specific constraints in data collection and monitoring, requiring tailored methodological approaches (Kulonen et al., 2019). Realising these recommendations will require coordinated action across multiple governance levels. Regional authorities in Buryatia and Zabaikalsky Krai, together with their counterparts in neighbouring Chinese provinces, should lead the development of joint early warning systems for forest fires. National statistical agencies—Rosstat, China's National Bureau of Statistics, and Mongolia's National Statistics Office—need to establish data sharing frameworks and harmonised indicator definitions, potentially under the auspices of BRICS or the Shanghai Cooperation Organisation. Trilateral government commissions could oversee the long-term establishment of demonstration zones, with international financial institutions providing funding for green infrastructure projects.

Several limitations should be acknowledged. First, the empirical analysis is limited to two Russian regions and does not include comparable statistical data from Chinese or Mongolian border territories due to data comparability constraints. Second, the high volatility of forest fire data depending on weather conditions means that the extreme values observed in 2024 may not represent a permanent trend. Third, subjective indicators (e.g., perceived quality of life, satisfaction with environmental conditions) are absent from the analysis. Fourth, the study does not employ econometric modelling to quantify causal relationships between SDG indicators. Future research should address these limitations by expanding the comparative base to include the Inner Mongolia Autonomous Region and Mongolia, incorporating sociological tools, and developing spatial econometric models to measure transboundary spillover effects.

6. Conclusions

The comprehensive analysis of official statistical data for 2010–2024, encompassing 39 indicators across six

Sustainable Development Goals (SDGs 6, 8, 11, 13, 15, and 17), reveals that the Republic of Buryatia and Zabaikalsky Krai, despite sharing a border with China and Mongolia and facing similar environmental and socio-economic constraints, have followed markedly divergent trajectories of sustainable development. These contrasting patterns, together with the common transboundary challenges identified, lead to several key conclusions that carry both scientific and policy-relevant implications.

Zabaikalsky Krai demonstrates above-average economic performance (SDG 8) but is rated as Critically below in water and sanitation (SDG 6) and sustainable cities (SDG 11), as well as Below in life on land (SDG 15). The Republic of Buryatia, by contrast, performs Above the national average in life on land (SDG 15) and partnerships (SDG 17), shows Mixed results in water and sanitation (SDG 6) and sustainable cities (SDG 11), and lags economically with a Below rating for SDG 8. Both regions share Extreme vulnerability under climate action (SDG 13) due to unprecedented forest fire areas in 2024, and both dispose of almost all municipal solid waste in landfills without recycling.

Several policy implications follow from these findings. The findings support the need for coordinated actions at different time horizons. In the short-term, joint Russia-China early warning systems for forest fires and harmonisation of environmental standards for transboundary water bodies (the Selenga basin) should be prioritised. In the medium-term, institutional arrangements should focus on agreed monitoring methodologies and data sharing protocols. In the long-term, the establishment of trilateral (Russia-China-Mongolia) sustainable development demonstration zones—building on existing transboundary reserves such as Dauria—could serve as platforms for testing integrated approaches to green growth, circular economy, and low-carbon development.

The study has several limitations that should be acknowledged. The empirical analysis is limited to two Russian regions and does not include comparable statistical data from Chinese or Mongolian border territories due to data comparability constraints. The high volatility of forest fire data means that the extreme values observed in 2024 may not represent a permanent trend. Subjective indicators (perceived quality of life, satisfaction with environmental conditions) are absent from the analysis, and the study does not employ econometric modelling to quantify causal relationships between SDG indicators.

Directions for future research emerge from these limitations. Further work should expand the comparative base to include the Inner Mongolia Autonomous Region and Mongolia, incorporating official statistical data from China's National Bureau of Statistics and Mongolia's National Statistics Office once comparability issues are resolved. Sociological tools should be employed to capture subjective dimensions such as perceived quality of life and satisfaction with environmental conditions. Spatial econometric models need to be developed to quantify transboundary spillover effects, particularly for forest fire propagation and water pollution in the Selenga basin. Finally, the harmonisation of statistical systems within BRICS and the Shanghai Cooperation Organisation remains a priority for enabling robust cross-national SDG assessments, and the proposed trilateral demonstration zones (Russia-China-Mongolia) could serve as testing grounds for agreed indicator frameworks.

Author Contributions

Conceptualization, N.L. and A.M.; methodology, N.L.; formal analysis, N.L. and A.M.; investigation, N.L. and A.M.; resources, N.L.; data curation, A.M.; writing—original draft preparation, N.L.; writing—review and editing, N.L. and A.M.; visualization, N.L.; supervision, N.L.; project administration, A.M.; funding acquisition, A.M. All authors have read and agreed to the published version of the manuscript.

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Data Availability

The data used to support the research findings are available from the corresponding author upon request. The official statistical data are publicly accessible through the Federal State Statistics Service of the Russian Federation (Rosstat) at <https://rosstat.gov.ru>.

Conflicts of Interest

The authors declare no conflicts of interest.

References

Alsaif, T. M., Ibrahim, U. M., Al-Hamad, I. A., & Abdulkader, R. A. (2025). Competitive advantages of Ha'il in

- the light of 2030 vision: Managing the economic ecosystem according to sustainable strategic management. *Environ. Dev. Sustain.*, 27(10), 25499–25523. <https://doi.org/10.1007/s10668-024-04566-w>.
- Atanov, N. I. & Munkodugarova, A. B. (2016). Economic dynamics in Russia's Transbaikal and Mongolia. *Probl. Econ. Transit.*, 58(7–9), 805–820. <https://doi.org/10.1080/10611991.2016.1251226>.
- Bataeva, B. S. & Kozhevina, O. V. (2020). Development of conditions of innovations institutionalization contributing to the Russian sustainable development. In *Lecture Notes in Networks and Systems* (pp. 48–53). Springer International Publishing. https://doi.org/10.1007/978-3-030-60929-0_7.
- Bychkov, I. V. & Orlova, I. I. (2018). Formation of effective cooperation instruments for development projects in the Lake Baikal basin. *IOP Conf. Ser.: Earth Environ. Sci.*, 190, 012047. <https://doi.org/10.1088/1755-1315/190/1/012047>.
- Dong, S., Li, Y., Li, Z., Li, F., Cheng, H., Yang, Y., Bilgaev, A., Zheng, J., & Bazarzhapov, T. (2018). Ecological environment risks and green development modes of China-Mongolia-Russia economic corridor. *IOP Conf. Ser.: Earth Environ. Sci.*, 190, 012053. <https://doi.org/10.1088/1755-1315/190/1/012053>.
- Dong, S., Zheng, J., Li, Y., Li, Z., Li, F., Jin, L., Yang, Y., & Bilgaev, A. (2019). Quantitative analysis of the coupling coordination degree between urbanization and eco-environment in Mongolia. *Chin. Geogr. Sci.*, 29(5), 861–871. <https://doi.org/10.1007/s11769-019-1074-7>.
- Ferrer-Roca, N., Guia, J., & Blasco, D. (2020). Partnerships and the SDGs in a cross-border destination: The case of the Cerdanya Valley. *J. Sustain. Tour.*, 30(10), 2410–2427. <https://doi.org/10.1080/09669582.2020.1847126>.
- Filin, S., Yakushev, A., Berdikulov, M., Velikorossov, V., & Chaikovskiy, A. (2020). Prospects of the Russian-Mongolian interaction in the innovative sector of agriculture. *E3S Web Conf.*, 176, 03010. <https://doi.org/10.1051/e3sconf/202017603010>.
- Hilty, J. A., Keeley, A. T. H., Lidicker, W. Z., & Merenlender, A. M. (2019). *Corridor Ecology: Linking Landscapes for Biodiversity Conservation and Climate Adaptation* (2nd ed.). Princeton University Press, Island Press. <https://doi.org/10.2307/jj.40831460>.
- Intergovernmental Panel on Climate Change. (2023). *Climate Change 2022—Impacts, Adaptation and Vulnerability: Working Group II Contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. Cambridge University Press. <https://doi.org/10.1017/9781009325844>.
- Kulonen, A., Adler, C., Bracher, C., & Dach, S. W. V. (2019). Spatial context matters in monitoring and reporting on Sustainable Development Goals: Reflections based on research in mountain regions. *GAIA—Ecol. Perspect. Sci. Soc.*, 28(2), 90–94. <https://doi.org/10.14512/gaia.28.2.5>.
- Medeiros, E. (2020). The territorial dimension of the United Nations Sustainable Development Goals. *Area*, 53(2), 292–302. <https://doi.org/10.1111/area.12681>.
- Nilsson, M., Chisholm, E., Griggs, D., Howden-Chapman, P., McCollum, D., Messerli, P., Neumann, B., Stevance, A. S., Visbeck, M., & Stafford-Smith, M. (2018). Mapping interactions between the sustainable development goals: Lessons learned and ways forward. *Sustain. Sci.*, 13(6), 1489–1503. <https://doi.org/10.1007/s11625-018-0604-z>.
- Nilsson, M., Griggs, D., & Visbeck, M. (2016). Policy: Map the interactions between Sustainable Development Goals. *Nature*, 534, 320–322. <https://doi.org/10.1038/534320a>.
- Siddiqui, A., Altekari, S., Kautish, P., Fulzele, S., Kulkarni, N., Siddiqui, M., & Bashir, M. F. (2023). Review of measurement of sustainable development goals: A comprehensive bibliometric and visualized analysis. *Environ. Sci. Pollut. Res.*, 30, 91761–91779. <https://doi.org/10.1007/s11356-023-28887-x>.
- Tyrins, A. N. & Vasilyeva, E. V. (2020). Sustainable development indicators in benchmarks of Russia's regional policy. *IOP Conf. Ser.: Earth Environ. Sci.*, 534, 012003. <https://doi.org/10.1088/1755-1315/534/1/012003>.
- United Nations. (1992). *Rio Declaration on Environment and Development*. Routledge.
- United Nations. (2015). *Transforming our World: The 2030 Agenda for Sustainable Development*.
- Violin, S. I. (2018). Development of the transboundary economic cooperation within the China-Mongolia-Russia economic corridor. *IOP Conf. Ser.: Earth Environ. Sci.*, 190, 012065. <https://doi.org/10.1088/1755-1315/190/1/012065>.
- Wang, P., Li, J., Hou, J., & Sun, W. (2023). Innovation demonstration zones for sustainable development agenda: A national pilot practice in China. *Environ. Dev. Sustain.*, 27(1), 141–156. <https://doi.org/10.1007/s10668-023-03906-6>.
- Wang, Y., Yuan, J., & Lu, Y. (2020). Constructing demonstration zones to promote the implementation of Sustainable Development Goals. *Geogr. Sustain.*, 1(1), 18–24. <https://doi.org/10.1016/j.geosus.2020.02.004>.
- Wei, C., Meng, J., Zhu, L., & Han, Z. (2023). Assessing progress towards sustainable development goals for Chinese urban land use: A new cloud model approach. *J. Environ. Manag.*, 326, 116826. <https://doi.org/10.1016/j.jenvman.2022.116826>.
- World Commission on Environment & Development. (1987). *Our Common Future*. Oxford University Press. <http://un-documents.net/our-common-future.pdf>
- Xie, H., Wen, J., & Choi, Y. (2021). How the SDGs are implemented in China—A comparative study based on

- the perspective of policy instruments. *J. Clean. Prod.*, 291, 125937. <https://doi.org/10.1016/j.jclepro.2021.125937>.
- Zeitoun, M. & Warner, J. (2006). Hydro-hegemony: A framework for analysis of trans-boundary water conflicts. *Water Policy*, 8(5), 435–460. <https://doi.org/10.2166/wp.2006.054>.
- Zhu, J. (2017). The 2030 agenda for sustainable development and China's implementation. *Chin. J. Popul. Resour. Environ.*, 15(2), 142–146. <https://doi.org/10.1080/10042857.2017.1322864>.
- Zhu, J., Sun, X., He, Z., & Zhao, M. (2018). Are SDGs suitable for China's sustainable development assessment? An application and amendment of the SDGs Indicators in China. *Chin. J. Popul. Resour. Environ.*, 17(1), 25–38. <https://doi.org/10.1080/10042857.2018.1544753>.